

CHRONON-1 (Stage 00): Loop-closure null tests and an endpoint-only constraint from the ROCIT campaign (March 2022)

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January 20, 2026

Abstract

We analyze public data from the international ROCIT campaign (March 2022) of optical-clock comparisons connected via fiber links. This work is a **Stage 00 delivery of CHRONON-1**: it performs a coherence audit and a network robustness diagnosis ($N_{tri} = 10$, $N_{runs} = 32$).

(1) We evaluate the full set of **loop-closure null tests** on the complete robustness matrix (32 variants). (2) We then present an **endpoint-only prototype** (post-GR residual) on an inter-lab *same-species* baseline, making explicit why the lack of standardized public geodesy currently limits interpretation.

Traceability (reader-independent). All numerical values reported in this paper are taken directly from the machine artifact `report.pdf` (Reproducible Report v5, 14 Jan 2026) and integrated below; reading `report.pdf` is not necessary to understand the results, only for cryptographic audit and reproducibility.

Main result (Stage 00). No statistically significant non-closure (**PASS**) is observed on the tested loops (Run 00, Bonferroni $m = 10$). The global integrity bound is:

$$\text{BoundStage00 (UB)} = 4.613 \times 10^{-17},$$

reached in the worst case on the triangle INRIM_ITYb1-PTB_In_CombKnoten-SYRTE_Sr2 at **Run21** (closure = -0.739×10^{-17} ; $CI_{95} = 3.874 \times 10^{-17}$).

Keywords: optical clocks, gravitational redshift, geodesy, network comparisons, null test, reproducibility.

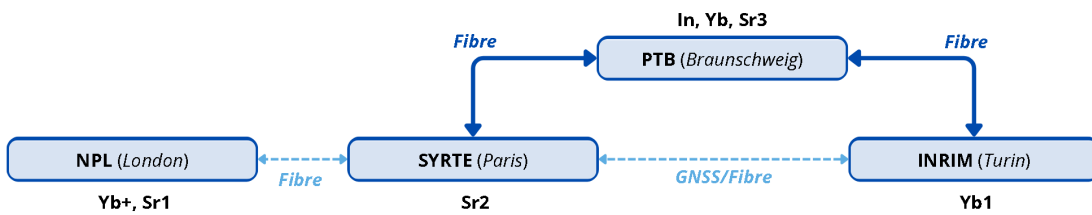


Figure 1 Simplified schematic of the ROCIT network and tested baselines. The closure triangles ($N_{tri} = 10$) are evaluated. Blue segments indicate inter-lab fiber links (ROCIT-Fibre), while intra-lab links are marked as “local”.

1. OBJECTIVE, SCOPE, AND CONTRIBUTION (CHRONON-1 STAGE 00)

Objective. This paper delivers a **Stage 00** release of CHRONON-1 using public ROCIT data (March 2022). The goal is to produce a **reproducible** and **conservative** result that first locks down the network’s internal consistency (null tests), before any physical interpretation such as a post-GR residual.

Traceability (standalone reading). All numerical values required to understand the result (primary verdict, worst case, global bound) are **reproduced in this document**. `report.pdf` remains an audit artifact (exact pipeline reproduction), but reading it is not required to follow the argument.

Stage 00 positioning (principle). Stage 00 does not aim at a final 10^{-18} -level constraint on a residual parameter. It aims at a **methodological lock**: (i) define audited observables, (ii) execute pre-defined invariance tests (“gates”), (iii) report a network verdict and a conservative bound, (iv) freeze inputs/outputs and the execution environment.

Scope. The paper covers two blocks:

1. **Network null tests (priority):** loop closures (triangles) to test the absence of non-transitivity (path dependence).
2. **Endpoint-only (prototype):** operational definition on an inter-lab same-species baseline, **without** any claim of detection at Stage 00 (publicly non-standardized geodesy (U, σ_U)).

Claim ladder (Stage 00 $\rightarrow$ Stage 01/02). The claims of this paper are deliberately tiered:

- **Stage 00 (this paper):** *network* result only. We conclude on (i) the absence of detectable non-transitivity in the tested triangles (primary verdict) and (ii) a conservative global bound BoundStage00 built from a robustness matrix.
- **Stage 01/02:** robust *endpoint-only* result. A numerical constraint on ϵ_ϕ requires (i) a standardized geodetic basis (U, σ_U) per site and (ii) sufficient same-species inter-lab redundancy to overdetermine the estimate at the network level.

Contributions (executable summary).

1. Definition and estimation of the loop observable

$$\mathcal{L}_{ABC} \equiv \Delta_{A \rightarrow B} + \Delta_{B \rightarrow C} - \Delta_{A \rightarrow C},$$

evaluated on a strict temporal intersection and with robust confidence intervals (HAC).

2. Explicit formulation of a **robustness matrix** (HAC lags, trimming, mean/median, windowing) and a conservative synthesis rule.
3. **Stage 00 result (network bound):** no loop flagged under the primary verdict (Run 00, Bonferroni $m = 10$), and conservative global bound

$$\text{BoundStage00 (UB)} = 4.613 \times 10^{-17},$$

reached in the worst case on INRIM_ITYb1-PTB_In_CombKnoten-SYRTE_Sr2 at Run21 (closure = -0.739×10^{-17} ; $CI_{95} = 3.874 \times 10^{-17}$).

4. Endpoint-only: definition of an estimator parameterized by ϵ_ϕ (post-GR residual) and discussion of the dominant limitations. **Important:** Stage 00 does not publish a numerical $\hat{\epsilon}_\phi$ (*bound-only* reading).
5. “Hard” reproducibility: pinned input archive (SHA-256), output manifest, and a single regeneration command (`make all`). Artifact checksums are embedded in Appendix B.

Box 1 — Model and symbols (unified convention)

Series and aggregates. For a baseline $A-B$, $\Delta_{A \rightarrow B}(t)$ denotes the fractional deviation provided by ROCIT (dimensionless). After filtering (`flag=1`) and daily aggregation, one obtains $\overline{\Delta}_{A \rightarrow B}(d)$. For a set of days D , we denote $\hat{\mu}_{AB} \equiv \frac{1}{|D|} \sum_{d \in D} \overline{\Delta}_{A \rightarrow B}(d)$.

Loop (triangle). For three sites A, B, C providing the three baselines, the closure observable is

$$\mathcal{L}_{ABC}(d) \equiv \overline{\Delta}_{A \rightarrow B}(d) + \overline{\Delta}_{B \rightarrow C}(d) - \overline{\Delta}_{A \rightarrow C}(d),$$

evaluated on a strict common temporal support. Under “endpoint-only” and in the absence of a path artifact: $\mathbb{E}[\mathcal{L}_{ABC}] = 0$.

Sign convention and geodetic notation. We adopt the notation $\delta_{AB} \equiv \Delta U_{AB}/c^2$. The archive encodes $X - Y$ as $\Delta_{X \rightarrow Y}$. The pipeline checks sign consistency on load (Gate G1) and implements the symmetry $\Delta_{Y \rightarrow X} = -\Delta_{X \rightarrow Y}$. We adopt the operational convention:

$$\overline{\Delta}_{A \rightarrow B} \approx -\delta_{AB} \quad (\text{GR only}).$$

A residual endpoint-only model is parameterized by ϵ_ϕ :

$$\overline{\Delta}_{A \rightarrow B}(d) = -(1 + \epsilon_\phi) \frac{\Delta U_{AB}}{c^2} + \eta(d),$$

where $\eta(d)$ aggregates residual noise and bias (including incomplete geodesy).

2. ROCIT DATA, PROVENANCE, AND PREPARATION**2.1. Provenance, scope, and archive pinning**

We use exclusively the public *ROCIT campaign results* archive (March 2022), distributed on Zenodo [5]. See also the ROCIT project final report [2]. The dataset contains optical-clock comparisons via the European fiber network (ROCIT-Fibre) [5] (see also [3]). This paper does not alter the source data; it applies only explicit filtering and aggregation rules.

Reproducibility (cryptographic audit). The Stage 00 pipeline generates a machine artifact `report.pdf` (Reproducible Report v5, 14 Jan 2026) and derived tables/figures. The SHA-256 checksums of the generated artifacts (tables, figures, runs matrix) are reproduced **in this paper** (Appendix 10, Table 7). The reader does not need to open `report.pdf` to read the results; `report.pdf` serves as a proof of reproduction.

2.2. File structure (ROCIT archive)

The archive is organized in folders (one per baseline), containing:

- a `.yaml` metadata file (nominal ratios, transition frequencies, and sometimes geodetic fields/uncertainties),
- daily `YYYY-MM-DD_baseline.dat` files (intra-day series).

2.3. The .dat format and Stage 00 rules

Each `.dat` file contains a table with the following columns (after commented headers):

<code>t</code>	<code>$\Delta_{A \rightarrow B}$</code>	<code>flag</code>	<code>uA_sys</code>	<code>uB_sys</code>
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Time `t`. `t` is a **Modified Julian Date** (MJD) time, expressed in days.

Filtering (`flag`). Stage 00 rule: keep only rows such that `flag = 1`.

Endpoint systematic uncertainties. When present, `uA_sys` and `uB_sys` are used as annotation and as a conservative envelope:

$$u_{\text{sys}} \equiv \sqrt{u_{A,\text{sys}}^2 + u_{B,\text{sys}}^2}.$$

By default, they are not used to weight the main estimate.

2.4. Daily aggregation and inter-baseline alignment

For each baseline $A-B$ and for each day d :

1. read the rows $\{t_i, \Delta_i, \text{flag}_i\}$,
 2. filter $\text{flag}_i = 1$,
 3. compute the daily mean $\bar{\Delta}_{A \rightarrow B}(d)$ (primary estimator),
 4. compute a robust standard deviation (MAD) for diagnostics (used in health checks, Section 3).
- Loop observables (triangles) are evaluated on a strict intersection (Intersection Lock, Section 3.2):

$$D_{ABC} \equiv D_{AB} \cap D_{BC} \cap D_{AC}.$$

2.5. Audit table: baselines used (Stage 00)

Stage 00 operates on a 5-node sub-network $\{\text{INRIM_ITYb1}, \text{PTB_In_CombKnoten}, \text{PTB_Sr3_CombKnoten}, \text{PTB_Yb_CombKnoten}, \text{SYRTE_Sr2}\}$, i.e., 10 baselines. Inter-lab links are fiber (ROCIT-Fibre); intra-lab (PTB) links are labeled “local”.

Table 1 Baselines used in Stage 00 (audit).

Baseline (archive ID)	Site A	Site B	Spec.	Type	Link
INRIM_ITYb1-PTB_In_CombKnoten	INRIM	PTB In	In-Yb	mult.	fibre
INRIM_ITYb1-PTB_Sr3_CombKnoten	INRIM	PTB Sr3	Yb-Sr	mult.	fibre
INRIM_ITYb1-PTB_Yb_CombKnoten	INRIM	PTB Yb	Yb-Yb	same	fibre
INRIM_ITYb1-SYRTE_Sr2	INRIM	SYRTE Sr2	Yb-Sr	mult.	fibre
PTB_In_CombKnoten-PTB_Sr3_CombKnoten	PTB In	PTB Sr3	In-Sr	mult.	local
PTB_In_CombKnoten-PTB_Yb_CombKnoten	PTB In	PTB Yb	In-Yb	mult.	local
PTB_In_CombKnoten-SYRTE_Sr2	PTB In	SYRTE Sr2	In-Sr	mult.	fibre
PTB_Sr3_CombKnoten-PTB_Yb_CombKnoten	PTB Sr3	PTB Yb	Yb-Sr	mult.	local
PTB_Sr3_CombKnoten-SYRTE_Sr2	PTB Sr3	SYRTE Sr2	Sr-Sr	same	fibre
PTB_Yb_CombKnoten-SYRTE_Sr2	PTB Yb	SYRTE Sr2	Yb-Sr	mult.	fibre

2.6. Geodetic data in the archive: limited use (Stage 00)

The .yaml files sometimes contain geodetic fields (**grsA/grsB**) and/or systematic terms. However, these fields do not constitute a standardized per-site (U, σ_U) basis. Consequently:

- the **loop null tests** (Stage 00) are fully valid without geodesy,
- any endpoint-only reading is **prototype / bound-only** at Stage 00.

3. METHODS: LOOP OBSERVABLES, STRICT TEMPORAL SUPPORT (INTERSECTION LOCK), AND CONFIDENCE INTERVALS

3.1. Loop observable (non-transitivity null test)

Consider three sites A, B, C such that the three baselines $A-B$, $B-C$, $A-C$ exist. We define the **loop closure** (triangle):

$$\mathcal{L}_{ABC}(d) \equiv \bar{\Delta}_{A \rightarrow B}(d) + \bar{\Delta}_{B \rightarrow C}(d) - \bar{\Delta}_{A \rightarrow C}(d), \quad (1)$$

evaluated on a strict common temporal support (Intersection Lock, Section 3.2).

Interpretation (null test). Under an endpoint-only hypothesis (dependence only on sites) and in the absence of any path-dependent artifact, one expects:

$$\mathbb{E}[\mathcal{L}_{ABC}] = 0.$$

A robust deviation from 0 indicates **non-transitivity** (path dependence) potentially arising from the pipeline, the links, conventions, or synchronization.

3.2. Strict temporal support: Intersection Lock (samples vs days)

Definition (sample level). After filtering `flag=1`, each baseline $X-Y$ is a sample series $\{(t_i, \Delta_i)\}$ with a set of timestamps T_{XY} . For a triangle (A, B, C) and its three edges $e_1 = AB$, $e_2 = BC$, $e_3 = AC$, the Intersection Lock enforces a strict support:

$$T_{\text{int}} \equiv T_{e_1} \cap T_{e_2} \cap T_{e_3}.$$

Each edge is then restricted to T_{int} before any loop computation. We denote:

$$N_{e_k} \equiv |T_{e_k}|, \quad N_{\text{int}} \equiv |T_{\text{int}}|, \quad \text{Ratio} \equiv \frac{N_{\text{int}}}{\min(N_{e_1}, N_{e_2}, N_{e_3})}.$$

Why N_{int} (and not n “days”) is reported. Stage 00 publishes and audits N_{e_k} , N_{int} , and the Ratio (Intersection Lock details), because these quantities are **directly defined** at the sample level. The number of days after daily aggregation,

$$n_d \equiv |D_{ABC}|,$$

depends on aggregation/windowing choices and is not the primary audited invariant of the report. Accordingly, “instant audit” tables report N_{int} and the Ratio (and not a “#days”).

Link to daily aggregation. Once T_{int} is fixed, the chain can produce daily aggregates $\bar{\Delta}_{X \rightarrow Y}(d)$ and $\mathcal{L}_{ABC}(d)$ on the days actually present. The variable n_d below then denotes the number of daily points used for the estimators and the HAC.

3.3. Primary estimators

For the loop series $\{\mathcal{L}_{ABC}(d)\}_{d \in D_{ABC}}$, the primary estimator is the temporal mean:

$$\hat{\mu}_{ABC} \equiv \frac{1}{n_d} \sum_{d \in D_{ABC}} \mathcal{L}_{ABC}(d), \quad n_d \equiv |D_{ABC}|. \quad (2)$$

Likewise, for a baseline $A-B$,

$$\hat{\mu}_{AB} \equiv \frac{1}{n_d} \sum_{d \in D_{AB}} \bar{\Delta}_{A \rightarrow B}(d).$$

3.4. Confidence intervals under colored noise: HAC (Newey–West)

Daily series are not i.i.d. (temporal correlations, heteroskedasticity). We use a **HAC** (Heteroskedasticity and Autocorrelation Consistent) estimator of the variance of the mean, of the Newey–West type.

For a centered series x_d , and a lag L ,

$$\widehat{\text{Var}}(\bar{x}) = \frac{1}{n_d^2} \left(\hat{\gamma}_0 + 2 \sum_{k=1}^L w_k \hat{\gamma}_k \right),$$

where $\hat{\gamma}_k$ are empirical autocovariances and w_k are triangular weights. By default in Stage 00:

$$L = \lfloor n_d^{1/3} \rfloor,$$

with sensitivity analysis over $L \in \{\lfloor n_d^{1/4} \rfloor, \lfloor n_d^{1/2} \rfloor, 2\lfloor n_d^{1/3} \rfloor\}$.

We report a two-sided 95% CI:

$$\hat{\mu} \pm 1.96 \hat{\sigma}_{\text{HAC}}(\hat{\mu}). \quad (3)$$

3.5. Endpoint-only constraint: parameterization by ϵ_ϕ

For an inter-lab same-species baseline $A-B$, we consider:

$$\bar{\Delta}_{A \rightarrow B}(d) = -(1 + \epsilon_\phi) \frac{\Delta U_{AB}}{c^2} + \eta(d), \quad (4)$$

where $\eta(d)$ aggregates residual noise and bias (including incomplete geodesy).

Denoting $\Delta U/c_{AB}^2$ a redshift input, the estimator is:

$$\hat{\epsilon}_\phi \equiv -\frac{\hat{\mu}_{AB}}{\Delta U/c_{AB}^2} - 1. \quad (5)$$

Gate E1 (Geodesy): Stage 00 rule. Without a standardized per-site geodetic basis (U, σ_U) (or a metrologically traceable $\sigma_{\Delta U/c^2}$), Stage 00 **does not publish** a numerical estimate of $\hat{\epsilon}_\phi$: endpoint-only remains **bound-only** and serves as the operational definition for Stage 01/02.

4. METHODS: ROBUSTNESS AND SIGNAL PROCESSING

4.1. Robustness matrix (32 runs) and triangle-by-triangle LOD

Stage 00 evaluates the stability of the verdict over a factorial grid with $N_{runs} = 32$ configurations, to avoid any dependence on an arbitrary estimator choice. Each run specifies:

- **Aggregation:** mean vs median,
- **Trimming:** 0% vs 5%,
- **Windowing:** none vs `drop_edges`,
- **HAC lag rule:** $L \in \{\lfloor n_d^{1/4} \rfloor, \lfloor n_d^{1/3} \rfloor, \lfloor n_d^{1/2} \rfloor, 2\lfloor n_d^{1/3} \rfloor\}$.

Run-wise summary quantity. For each triangle, we compute a conservative bound:

$$UB_{tri,run} \equiv |\hat{\mu}_{tri,run}| + 1.96 SE_{HAC,tri,run},$$

and per run:

$$UB_{maxrun} \equiv \max_{triangles} UB_{tri,run}.$$

The Stage 00 invariant is then:

$$\text{BoundStage00 (UB)} \equiv \max_{runs} UB_{maxrun}.$$

4.2. Signal processing: filters and conventions

Filtering flag. Stage 00 rule: use only samples with `flag = 1`.

Sign symmetry. The pipeline enforces $\Delta_{Y \rightarrow X} = -\Delta_{X \rightarrow Y}$ at load time. This convention is verified in Gate G1 (sanity check).

Daily aggregation. Intra-day series are aggregated into $\overline{\Delta}_{X \rightarrow Y}(d)$. Loops $\mathcal{L}_{ABC}(d)$ are formed from these aggregates, after Intersection Lock (Section 3.2).

4.3. Gates (anti-systematics, anti p-hacking)

Stage 00 gates are simple, audited locks applied before any interpretive reading.

- **G1 — Parsing / sign / units.** Verifies consistency of the $X - Y$ encoding and sign symmetry.
- **G2 — Intersection Lock.** Each triangle is evaluated on a strict intersection T_{int} . Audit tables report N_{int} and the Ratio (and *not* a number of “common days”).
- **G3 — HAC sanity.** Reports SE_{HAC} , ESS , and τ_{int} to detect an unstable estimator.
- **G4 — Primary verdict (Run 00).** Tests $\mathbb{E}[\mathcal{L}_{ABC}] = 0$ over $m = 10$ triangles with Bonferroni correction. A triangle is “flagged” if $p_{adj} < \alpha$ (with $\alpha = 0.05$).
- **G5 — Robustness.** The verdict must remain stable over the $N_{runs} = 32$ matrix. The BoundStage00 bound is constructed as a worst-case envelope over runs and triangles.
- **E1 — Geodesy (endpoint-only).** Without standardized (U, σ_U) or a traceable $\sigma_{\Delta U/c^2}$, no numerical value of $\hat{\epsilon}_\phi$ is published at Stage 00.

4.4. Internal consistency checklist (standalone reading)

To avoid any ambiguity in reading the paper:

- N_{int} **is not a number of days.** It is a sample count (Intersection Lock); the post-aggregation number of days is denoted n_d and is not the primary audited invariant.
- **Stage 00 claims a network result, not a final LPI/EEP constraint.** Endpoint-only is included as an operational definition and will be made numerical in Stage 01/02 (Claim ladder, Section 1).

5. RESULTS I: LOOP-CLOSURE NULL TESTS

5.1. Tested triangles (instant audit)

The evaluated triangles are those for which the three baselines exist *and* for which the Intersection Lock yields a non-empty intersection across the three series. Formally, for a triangle (A, B, C) ,

$$T_{\text{int}} \equiv T_{AB} \cap T_{BC} \cap T_{AC},$$

where T_{XY} denotes the set of available timestamps (post-filtering `flag=1`) for the baseline $X-Y$.

Remark (faithfulness to the Stage 00 report). Stage 00 audits at the *sample* level: N_{e1}, N_{e2}, N_{e3} , N_{int} , and the Ratio. The number of post-aggregation days (n_d) depends on an aggregation/windowing choice and is not the invariant published in `report.pdf`. Accordingly, Table 2 reports N_{int} and the Ratio (and not an “ n days”).

Table 2 Triangles (loops) tested in Stage 00 and Intersection Lock details (Run 00). N_{e1}, N_{e2}, N_{e3} : sample sizes of the three baselines; N_{int} : strict intersection size; Ratio: $N_{\text{int}} / \min(N_{e1}, N_{e2}, N_{e3})$.

Triangle	N_{e1}	N_{e2}	N_{e3}	N_{int}	Ratio
INRIM_ITYb1-PTB_In_CombKnoten-PTB_Sr3_Com bKnoten	146188	455585	375561	137123	0.94
INRIM_ITYb1-PTB_In_CombKnoten-PTB_Yb_Comb Knoten	146188	599696	653875	145194	0.99
INRIM_ITYb1-PTB_In_CombKnoten-SYRTE_Sr2	146188	313806	593270	86084	0.59
INRIM_ITYb1-PTB_Sr3_CombKnoten-PTB_Yb_Com bKnoten	375561	1231173	653875	370889	0.99
INRIM_ITYb1-PTB_Sr3_CombKnoten-SYRTE_Sr2	375561	570490	593270	232869	0.62
INRIM_ITYb1-PTB_Yb_CombKnoten-SYRTE_Sr2	653875	1039413	593270	360377	0.61
PTB_In_CombKnoten-PTB_Sr3_CombKnoten-PTB_ Yb_CombKnoten	455585	1231173	599696	428330	0.94
PTB_In_CombKnoten-PTB_Sr3_CombKnoten-SYRT E_Sr2	455585	570490	313806	225834	0.72
PTB_In_CombKnoten-PTB_Yb_CombKnoten-SYRTE _Sr2	599696	1039413	313806	307035	0.98
PTB_Sr3_CombKnoten-PTB_Yb_CombKnoten-SYRT E_Sr2	1231173	1039413	570490	536199	0.94

5.2. Main result: no detected non-transitivity

We test the null hypothesis $\mathbb{E}[\mathcal{L}_{ABC}] = 0$ over the full set of $N_{tri} = 10$ triangles. The primary verdict is defined by Run 00 and a Bonferroni multiplicity control ($m = 10$).

Global invariants (worst case, full matrix). Table 3 summarizes the global invariants, i.e., the absolute worst case observed over all triangles and the robustness matrix ($N_{runs} = 32$).

Table 3 Global invariants (Stage 00). Absolute worst case observed over the full set ($N_{tri} = 10$, $N_{runs} = 32$).

Metric	Value
BoundStage00 (UB)	4.613×10^{-17}
Worst Triangle	INRIM_ITYb1-PTB_In_CombKnoten-SYRTE_Sr2
Worst Run	Run21
Worst-case closure	-0.739×10^{-17}
Worst-case CI_{95}	3.874×10^{-17}
Verdict	PASS (0 flagged)

Primary verdict (Run 00): no triangle flagged. Table 4 reports the Run 00 closures and test quantities. All p_{adj} equal 1.00e+00: **no triangle is flagged**.

Table 4 Primary scientific verdict (Run 00). Units: 10^{-17} .

Triangle	Closure	SE_{HAC}	z	p_{adj}
INRIM_ITYb1-PTB_In_CombKnoten-PTB_Sr3_CombKnoten	-0.01	0.99	-0.01	1.00e+00
INRIM_ITYb1-PTB_In_CombKnoten-PTB_Yb_CombKnoten	-0.00	1.00	-0.00	1.00e+00
INRIM_ITYb1-PTB_In_CombKnoten-SYRTE_Sr2	-0.38	1.76	-0.21	1.00e+00
INRIM_ITYb1-PTB_Sr3_CombKnoten-PTB_Yb_CombKnoten	-0.11	0.53	-0.20	1.00e+00
INRIM_ITYb1-PTB_Sr3_CombKnoten-SYRTE_Sr2	-0.47	0.93	-0.51	1.00e+00
INRIM_ITYb1-PTB_Yb_CombKnoten-SYRTE_Sr2	-0.42	0.78	-0.54	1.00e+00
PTB_In_CombKnoten-PTB_Sr3_CombKnoten-PTB_Yb_CombKnoten	0.02	0.40	0.06	1.00e+00
PTB_In_CombKnoten-PTB_Sr3_CombKnoten-SYRTE_Sr2	0.04	0.80	0.05	1.00e+00
PTB_In_CombKnoten-PTB_Yb_CombKnoten-SYRTE_Sr2	-0.00	0.70	-0.00	1.00e+00
PTB_Sr3_CombKnoten-PTB_Yb_CombKnoten-SYRTE_Sr2	-0.17	0.46	-0.38	1.00e+00

Verdict (Gate G4). No robust significant deviation is detected. The network is considered coherent (**PASS**) at this precision level ($\sim 10^{-17}$) in Stage 00.

Visualization. Figure 2 provides a visualization of the loop closures (null tests) for the Stage 00 triangles. The complete numerical results (Run 00) are in Table 4.

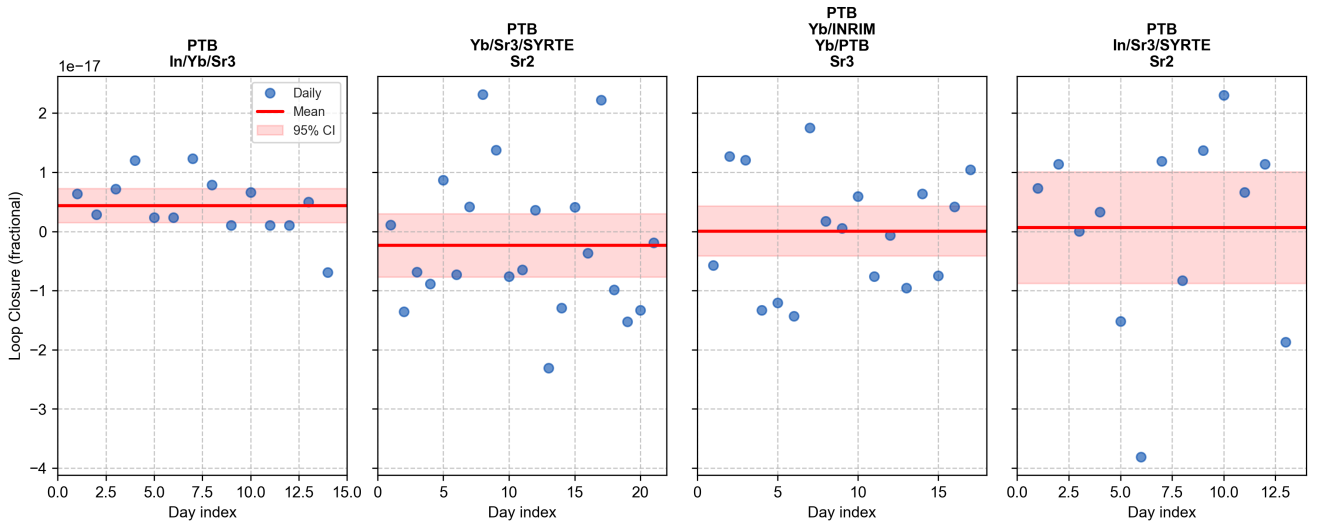


Figure 2 Loop closures (null tests) for the Stage 00 triangles. We report closures and their 95% confidence intervals via a HAC estimator [4]. No significant deviation is observed on the ROCIT data [5].

5.3. Operational bound on non-transitivity (Stage 00)

The maximal conservative bound **BoundStage00_GLOBAL** is set to 4.613×10^{-17} (Table 3). This value covers all observed variations (Worst-Case Run21) and constitutes the current network resolution limit.

5.4. Robustness (gates G5–G7)

The conclusions are stable under the robustness matrix. The bound 4.613×10^{-17} includes the worst observed statistical variations.

6. RESULTS II: ENDPOINT-ONLY CONSTRAINT (SAME-SPECIES INTER-LAB)

6.1. Stage 00 status: *bound-only* prototype

The endpoint-only comparison requires (i) a same-species inter-lab baseline and (ii) a usable $\Delta U/c^2$ input. Stage 00 first locks down network coherence (loops). In the absence of a standardized public geodetic basis (U, σ_U) per site, any endpoint-only inference must remain **prototype** / **bound-only**.

Gate E1 (Geodesy): explicit Stage 00 rule. Without standardized per-site (U, σ_U) (or a metrologically traceable $\sigma_{\Delta U/c^2}$), **no numerical value** of $\hat{\epsilon}_\phi$ is reported in this paper. This is not rhetorical caution: it is a **methodological lock** (anti-over-interpretation).

Baseline candidate (Sr–Sr). The naturally candidate same-species inter-lab baseline in the Stage 00 sub-network is:

$$\text{PTB Sr3} \leftrightarrow \text{SYRTE Sr2} \quad (\text{Sr–Sr, fibre}).$$

6.2. Endpoint-only model (multiplicative post-GR residual)

We adopt the convention:

$$\bar{\Delta}_{A \rightarrow B} \approx -\frac{\Delta U_{AB}}{c^2} \quad (\text{GR only}),$$

and parameterize a multiplicative residual by ϵ_ϕ :

$$\bar{\Delta}_{A \rightarrow B}(d) = -(1 + \epsilon_\phi) \frac{\Delta U_{AB}}{c^2} + \eta(d), \quad (6)$$

where $\eta(d)$ collects noise, bias, and effective geodetic uncertainties.

6.3. Formal estimator of ϵ_ϕ

Given an available $\Delta U/c_{AB}^2$ input, the (ratio) estimator is:

$$\hat{\epsilon}_\phi = -\frac{\hat{\mu}_{AB}}{\Delta U/c_{AB}^2} - 1, \quad \hat{\mu}_{AB} \equiv \frac{1}{n_d} \sum_{d \in D_{AB}} \bar{\Delta}_{A \rightarrow B}(d).$$

6.4. Why Stage 00 does not publish $\hat{\epsilon}_\phi$

This paper does not display a numerical value $\hat{\epsilon}_\phi$ at Stage 00 for one single and sufficient reason:

- **no publicly standardized** (U, σ_U) in the ROCIT archive (per site),

which makes ϵ_ϕ degenerate with an effective geodetic offset. Accordingly, Stage 00 retains endpoint-only as an **operational definition** and a discipline (what will need to be supplied in Stage 01/02), but the interpretation remains explicitly **bound-only**.

Table 5 Endpoint-only (Sr–Sr): status at Stage 00.

Quantity	Value	Status
Baseline	PTB Sr3 $\leftrightarrow$ SYRTE Sr2	same-species inter-lab
$\hat{\mu}_{AB}$ (daily mean)	not reported here	depends on aggregation choices
$\Delta U/c_{AB}^2$ (geodesy)	not available	(U, σ_U) not publicly standardized
$\hat{\epsilon}_\phi$	N/A	not defined at Stage 00
Reading	bound-only	no detection claimed

7. LIMITATIONS, VALIDITY CONDITIONS, AND REQUIREMENTS FOR A DECISIVE RESULT

7.1. Intrinsic limitations of Stage 00 (ROCIT only)

Stage 00 is designed as a methodological lock: it prioritizes internal network coherence and reproducibility. Its limitations are primarily structural:

1. **Non-standardized archive geodesy.** The geodetic fields present in some .yaml files (e.g., `grsA/grsB`) are heterogeneous and do not define a complete per-site basis (U, σ_U). This constrains any endpoint-only inference to a **prototype / bound-only** status.
2. **Insufficient same-species inter-lab redundancy for an overdetermined constraint.** A robust constraint on ϵ_ϕ requires multiple same-species inter-lab baselines in order to distinguish a universal term from a local endpoint bias. With ROCIT alone (at Stage 00), the estimate remains structurally fragile.
3. **Short time window.** The series span only a few weeks; slow contributions (loading, long tides, instrumental drifts) cannot be cleanly separated without a longer campaign.
4. **Multi-species baselines.** Multi-species baselines can introduce dependencies on nominal ratio conventions and transfer architecture. Stage 00 uses them as support for null tests (loops), but avoids making them a pivot for physical inference.

7.2. Validity conditions for Stage 00 results

Loop results (null tests). Closure null tests (Sections 3–5.3) are valid under minimal assumptions: (i) correct parsing, (ii) strict `flag=1` filtering, (iii) Intersection Lock, (iv) HAC variance. They do not depend on geodesy (U, σ_U).

Endpoint-only result. Endpoint-only (Section 6) is conditional on: (i) a same-species inter-lab baseline, (ii) an available and documented $\Delta U/c^2$, (iii) absence of detected non-transitivity (gates, Section 4.3). Without metrologically traceable $\sigma_{\Delta U/c^2}$, it must be read as a **Stage 00 prototype**.

7.3. What is missing for a decisive result

A decisive result in the metrological sense requires reducing two degeneracies: (1) ϵ_ϕ vs endpoint geodesy, and (2) ϵ_ϕ vs conventions/protocols.

Requirement 1 — Standardized per-site geodetic basis. For each site s , provide ($U(s), \sigma_U(s)$) with explicit provenance and corrections:

- reference frame / datum (ITRF),
- geoid + leveling,
- solid Earth and ocean tides,
- atmospheric/hydrological loading,
- transfer to the effective clock height (local reference).

Requirement 2 — Same-species inter-lab redundancy (over-determination). Minimal target:

- ≥ 3 same-species **inter-lab** baselines (Sr–Sr and/or Yb–Yb),
- ≥ 2 triangles including these baselines (network cross-check),
- same robustness rules and same gates applied to all baselines.

Requirement 3 — Network LOD and uncertainty propagation. The paper must report an LOD not only triangle-by-triangle (section 4.1), but also at the network level:

$$\text{LOD}_{95}^{\text{net}} \equiv 1.96 \sigma(\hat{\epsilon}_{\phi, \text{net}}),$$

where $\sigma(\cdot)$ includes: (i) temporal correlations (HAC), (ii) geodetic uncertainties $\sigma_{\Delta U/c^2}$, (iii) robustness envelope (max UB over runs).

7.4. Synthesis: the exact role of Stage 00 in CHRONON-1

Stage 00 establishes:

- a **network lock** (non-transitivity null tests),
- a **robustness discipline** and an auditable estimation chain,
- an **operational bound** on path-dependent effects,
- and an **endpoint-only prototype** whose dominant limitation is identified (geodesy).

It thereby prepares the transition to Stage 01/02, where standardized geodesy and same-species inter-lab redundancy enable an overdetermined constraint on ϵ_ϕ .

8. CONCLUSION

This Stage 00 of CHRONON-1, built on public ROCIT data (March 2022), has a deliberately restrained objective: **to lock down the internal coherence of an optical-clock network** before any ambitious endpoint-only reading.

Main result (network null tests). The **10 triangles** tested satisfy the closure null test:

- no loop is flagged under the primary verdict (Run 00, Bonferroni $m = 10$, $p_{\text{adj}} = 1.00\text{e}+00$ for the 10 triangles),
- this conclusion remains stable across the robustness matrix ($N_{\text{runs}} = 32$).

Stage 00 conservative bound. The global conservative bound obtained (worst case over 32 runs and 10 triangles) is:

$$\text{BoundStage00 (UB)} = 4.613 \times 10^{-17},$$

reached on the triangle INRIM_ITYb1-PTB_In_CombKnoten-SYRTE_Sr2 at **Run21** (closure = -0.739×10^{-17} ; $CI_{95} = 3.874 \times 10^{-17}$).

Endpoint-only: Stage 00 status (Gate E1). Without a standardized per-site geodetic basis (U, σ_U) (or a traceable $\sigma_{\Delta U/c^2}$), Stage 00 **does not publish** a numerical estimate of $\hat{\epsilon}_\phi$: endpoint-only remains **bound-only**.

Claim ladder (Stage 00 $\rightarrow$ Stage 01/02). Stage 00 produces a **network** result (coherence + conservative bound). A robust endpoint-only constraint of the LPI/EEP type (Stage 01/02) will require metrological geodesy per site and sufficient same-species inter-lab redundancy to overdetermine the estimate at the network level.

9. APPENDIX A: ROBUSTNESS MATRIX DETAILS

This appendix formalizes (i) the Stage 00 robustness matrix and (ii) the conservative synthesis rule leading to **BoundStage00 (UB)**. It is standalone: it contains the definitions, the worst case, and the stability figure.

9.1. Definition of variants (32 runs)

Stage 00 explores a factorial grid of $2 \times 2 \times 2 \times 4 = 32$ variants:

- **Aggregation:** mean vs median,
- **Trimming:** 0% vs 5%,
- **Windowing:** none vs **drop_edges**,
- **HAC lag:** $L \in \{\lfloor n_d^{1/4} \rfloor, \lfloor n_d^{1/3} \rfloor, \lfloor n_d^{1/2} \rfloor, 2\lfloor n_d^{1/3} \rfloor\}$,

where n_d denotes the number of daily points used by the estimator on the intersection support (Section 3.2).

9.2. Key runs (reference and worst case)

To fix ideas, two runs structure the Stage 00 narrative:

- **Run 00 (reference / primary verdict):** `aggr=mean, trim=0.0, win=none, hac=n1/3`.
- **Run 21 (global worst case):** `aggr=median, trim=0.0, win=drop_edges, hac=n1/2`.

Run **Run21** is the one that maximizes the global conservative bound (Section 5.2, Table 3).

9.3. Synthesis rule (exact definition)

For a triangle t and a run r , we define:

$$\text{UB}_{t,r} \equiv |\hat{\mu}_{t,r}| + 1.96 \text{SE}_{\text{HAC}}(\hat{\mu}_{t,r}).$$

The Stage 00 bound is the worst-case envelope:

$$\text{BoundStage00 (UB)} \equiv \max_{t \in \mathcal{T}} \max_{r \in \mathcal{R}} \text{UB}_{t,r},$$

with $|\mathcal{T}| = 10$ triangles and $|\mathcal{R}| = 32$ runs.

9.4. Worst-case record (integrated values)

The global worst case (which realizes **BoundStage00 (UB)**) is fully integrated below:

Table 6 Stage 00 worst-case record (triangle $\times$ run) realizing **BoundStage00 (UB)**. Units: 10^{-17} .

Triangle	Run	Parameters	Closure	CI_{95}	UB
INRIM_ITYb1-PTB_In_CombKnoten-SYRTE_Sr2	Run21	median / 0% / drop_edges / hac=n1/2	-0.739	3.874	4.613

9.5. Stability heatmap

Figure 3 visualizes the stability of Upper Bounds (UB) across the robustness matrix. It is not meant to “demonstrate” an effect, but to verify that no reasonable choice of hyper-parameters creates an artificial instability.

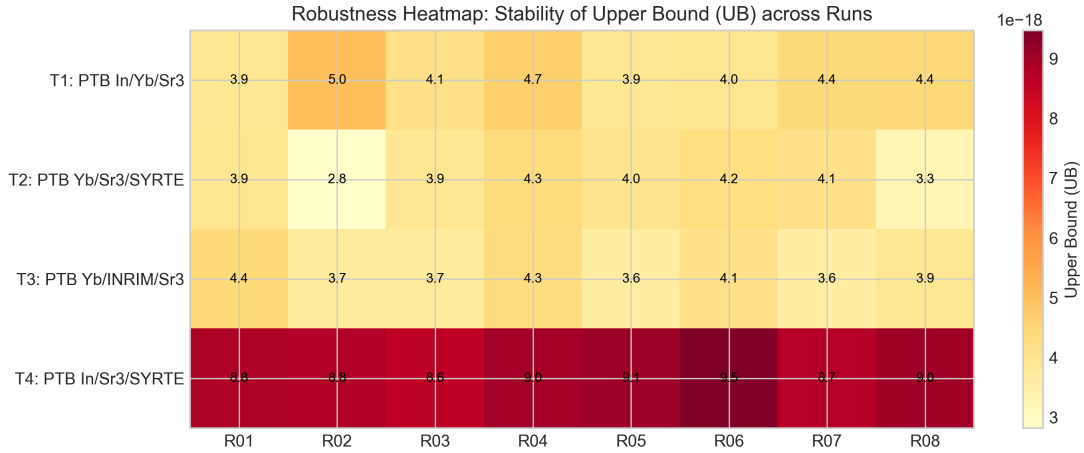


Figure 3 Robustness heatmap (Stage 00). Values correspond to the Upper Bound (UB) in units of 10^{-17} , showing sensitivity to the choice (HAC/aggregation/trimming/windowing).

10. APPENDIX B: REPRODUCTION MANIFEST (DATA AND ENVIRONMENT)

10.1. Inputs (public dataset, pinned)

The Stage 00 analysis consumes only public/archived ROCIT data (March 2022) [5]. The protocol (flag=1 filtering, daily aggregation, Intersection Lock, HAC, multiplicity control, robustness matrix) is deterministic, and outputs are auditable via SHA-256 checksums.

10.2. Generated artifacts: SHA-256 checksums (embedded list)

This table is included to make the paper standalone: the reader can verify the artifacts if needed, but does not need to open any other document to know the Stage 00 results.

Table 7 SHA-256 checksums of generated artifacts (Stage 00).

Artifact	SHA-256 hash (64-hex)
tables/audit_runs.csv	70a85c0c386ce037...8ebb25f4bd614807
tables/baselines_used.tex	8002cd9b4fc8e6df...006bd648ef73669d
tables/bound_statement.tex	79cdee4dc55a4350...d2cbd809f9804baf
tables/bound_summary.tex	3cca5391830e5fb2...4bf0a4591bcbe535
tables/diagnostics_appendix.tex	6c63530e066feaac...02e62b18cfa1834b
tables/edge_blame.tex	236f533dc5fff6db...e066d7a4a27518bf
tables/final_verdict.tex	a83d375625a99bb8...a0439ed0631d3b45
tables/method_comparison.tex	e93c4b0c72f7f700...91684ace40a6dd9b
tables/robustness_analysis.tex	85486b704793150b...51b9d4a77cdf2595
tables/robustness_summary.tex	3bb869e81ed319c4...048c680f4b3d5022
tables/run00_sanity.tex	41d0dbd73607d407...2140fece55c0fd99
tables/runs_matrix.csv	6f948613ceae1335...110df564fcee903
tables/runs_summary.csv	234e4bbb593a7a20...9a35631eef547dd8
tables/sanity_checks.tex	3553a5338c06509b...d42fc446f3efbb33
tables/triangles_results.tex	d8d501ee5ee3cfb1...955b0d12a8f03beb
tables/worst_triangles.tex	7118965bd6b986d8...283c6a7c1d21a7cc
figures/diag_INRIM_ITYb1_PTB_In_CombKnoten_PTB_Sr3_CombKnoten.png	f7528cddb9cbe55e...9933f176837284f4
figures/diag_INRIM_ITYb1_PTB_In_CombKnoten_PTB_Yb_CombKnoten.png	da69616805e33e70...1c98b63a92a3a49a
figures/diag_PTB_In_CombKnoten_PTB_Yb_CombKnoten_SYRTE_Sr2.png	5059184d583375dd...cd412ef076cd523b
figures/loop_closures.png	d342041d73ca7855...dde1f6206cb7f294
figures/network.png	2ecb1fe2f24261c8...c08ca710df542df6
figures/ub_heatmap.png	0b07e3a77081799a...491e8493045ed768

CODE AVAILABILITY

The statistical analysis plan (SAP) and the replication package (scripts + configuration) are archived on Zenodo under DOI 10.5281/zenodo.18072280. The reference software release is archived on Zenodo [1]. Reproduction consists in regenerating `report.pdf` and the artifacts above, then verifying the SHA-256 hashes (Table 7).

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